



Figure 2: Market size and use cases of artificial intelligence (Source: Statista)

Table 1: Key applications and use cases of AI in cybersecurity and network management

AI in cybersecurity and network forensics	AI-driven cyber deception
AI-driven cyber risk assessment	5G and IoT security
AI-driven SOC (security operations centre)	AI-enhanced penetration testing
AI-powered endpoint detection and response (EDR)	AI-powered firewalls and IDS/IPS
AI-powered SIEM optimisation	Adversarial AI detection
Anomaly detection	Automated data privacy compliance
Automated incident response	Automated patch management
Botnet detection	Cloud security automation
Cyber risk scoring	DNS security
Dark web monitoring	Deception technology
Deepfake detection	Digital forensics
Encrypted traffic analysis	Forensics log analysis
Fraud detection	Identity and access management (IAM)
Insider threat detection	Malware analysis
Network traffic analysis	Phishing detection
Predictive threat intelligence	Quantum-safe cryptography
Ransomware protection	Secure DevOps (DevSecOps)
Security information and event management (SIEM)	Supply chain security
Threat detection and prevention	Threat hunting
User behaviour analytics (UBA)	Voice and chatbot security
Zero trust security	Self-learning AI for adaptive security

AI in cybersecurity and network forensics

AI is not only useful for content writing but also for applications related to digital forensics and network management. It can be used for cyber forensics, network analytics and information security (Table 1).

A few AI-based open source tools that are popular for cyber forensics and network analytics are listed below.

Threat intelligence and SIEM (security information and event management) tools

Wazuh (<https://wazuh.com>)

Wazuh is an open source monitoring tool integrated with AI-powered endpoint security, log analysis and evaluations. It uses its extended detection and response (XDR) as well as security information and event management (SIEM) capabilities to offer a unified platform for diversified protection.

Its endpoint security features include malware detection, configuration assessment, and file integrity monitoring. Regulatory compliance, incident response and IT hygiene make up its security operations. It also offers log data analysis,